

Image Retrieval Using Pretrained CNN Embeddings: A Comparative Study of Similarity Metrics

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Abstract

This study implements and evaluates a content-based image retrieval (CBIR) system using deep feature representations from a pretrained Convolutional Neural Network (CNN). The framework addresses the semantic gap often present in traditional retrieval methods by using a pretrained ResNet-50 model, with its final classification layer removed, to extract 2048-dimensional feature embeddings from a small-scale proof-of-concept subset of 200 training images from the CIFAR-10 dataset. To enable efficient similarity search, these embeddings are indexed using the FAISS library. A key component of this research is an ablation study comparing the retrieval effectiveness of two fundamental similarity metrics: Euclidean (L2) distance and Cosine similarity. This comparison is critical for determining which metric better preserves the semantic relationships captured in the high-dimensional embedding space. A baseline retrieval system using traditional 3D Color Histograms is also implemented to provide a performance benchmark against handcrafted features. Quantitative evaluation on this small subset, measured by P@10 label consistency, compared all three methods: the CNN-Cosine method ($P@10 \approx 59.8\%$), the CNN-L2 method ($P@10 \approx 56.8\%$), and the Color Histogram baseline ($P@10 \approx 32.2\%$). Furthermore, the ablation study confirms that Cosine similarity provides more semantically accurate results than Euclidean distance. These findings are qualitatively supported by t-Distributed Stochastic Neighbour Embedding (t-SNE) visualizations, which show superior class separation and clustering when embeddings are normalized and compared using Cosine similarity. This work validates the efficacy of pretrained CNN embeddings for semantic retrieval and underscores the critical impact of metric selection in feature space.

1. Introduction

In the modern digital era, the proliferation of visual data is unprecedented. Billions of images are generated daily on social media, supported by surveillance systems, medical imaging devices, and personal archives. This explosion has created a critical need for efficient and effective methods to search, organize, and retrieve this data not just by metadata (such as filenames or tags) but by their actual visual content. This challenge is the foundation of Content-Based Image Retrieval (CBIR) [1].

Traditional CBIR systems have historically relied on handcrafted visual descriptors such as color histograms, texture features (e.g., Gabor filters), or local invariant features (e.g., SIFT, SURF) [2]. While effective for specific tasks like exact-duplicate detection or color-based matching, these methods suffer from a significant limitation known as the “semantic gap” [1]. This is the disparity between the low-level visual features (pixel colors, textures) that a computer can extract and the high-level semantic concepts (e.g., “dog,” “beach,” “car”) that a human user perceives and searches for.

The advent of deep learning, particularly Convolutional Neural Networks (CNNs), has provided a powerful solution to this problem. Trained on large-scale datasets such as ImageNet [3], deep CNNs learn a hierarchical representation of visual features, progressing from simple edges and textures in early layers to complex object parts and complete objects in deeper layers [4]. By leveraging transfer learning, one can use a pretrained CNN as a fixed feature extractor. The output vector from one of the network’s final layers—often referred to as a “deep embedding”—serves as a rich, high-dimensional descriptor that captures the complex semantic content of an image.

This project implements a complete CBIR pipeline based on this principle. We utilize a ResNet-50 [5] model, pretrained on ImageNet, to generate 2048-dimensional embeddings for images from the CIFAR-10 dataset [6]. For efficient retrieval, we employ the FAISS (Facebook AI Similar-

ity Search) library [7], a state-of-the-art framework for fast nearest-neighbor search in high-dimensional spaces. FAISS supports efficient indexing through both exact (brute-force) and approximate nearest-neighbor (ANN) approaches, enabling scalable image retrieval on large datasets.

The primary contributions of this work are as follows:

- **Implementation:** A complete CBIR pipeline is built using a pretrained ResNet-50 and the FAISS library.
- **Baseline Comparison:** The deep learning approach is benchmarked against a traditional 3D Color Histogram baseline to quantitatively demonstrate the superiority of semantic features over low-level descriptors.
- **Ablation Study:** A rigorous comparison between Euclidean (L2) distance and Cosine similarity is conducted to determine the optimal metric for comparing deep embeddings—a critical choice in system design.
- **Qualitative Analysis:** t-Distributed Stochastic Neighbour Embedding (t-SNE) [8] is used to visualize the embedding space, providing qualitative validation for the quantitative results and insights into how different metrics organize semantic concepts.

2. Related Work

The field of image retrieval has evolved significantly, moving from simple text-based metadata indexing to complex content-based analysis.

Traditional CBIR: Early CBIR systems focused on extracting global features. The most common of these is the Color Histogram [2], which quantizes the color distribution of an image into a fixed-size vector. While robust to scale and rotation, it completely disregards spatial information and semantic context. Other methods used texture descriptors or local invariant features such as SIFT (Scale-Invariant Feature Transform) [9]. To manage these local features, the Bag-of-Words (BoW) model, adapted from text retrieval, became popular [10]. In this approach, local descriptors are clustered to form a “visual vocabulary,” and an image is represented by a histogram of its visual word occurrences. Despite its success, the BoW model still relies on low-level features and struggles with the semantic gap.

Deep Learning for Feature Extraction: The paradigm shifted with the success of CNNs in the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) [3]. Models such as AlexNet [4], VGG [11], and ResNet [5] demonstrated an unparalleled ability to learn semantically rich feature hierarchies from data. Researchers quickly adapted these models for retrieval, repurposing them as feature extractors [12]. By feeding an image through a pretrained network and extracting the activation from a fully-connected layer (e.g., `fc7` in VGG or the `avgpool` layer in ResNet),

a dense, fixed-length vector (embedding) is produced. This vector serves as a powerful semantic signature for the image, where proximity in the embedding space corresponds to high visual and semantic similarity.

Similarity Metrics and Indexing: Given a query embedding, a CBIR system’s task is to find the nearest neighbors in a database of millions or billions of vectors. A naive linear scan, computing the distance to every vector, is computationally infeasible, which necessitates efficient indexing.

Furthermore, the choice of distance metric is critical. Euclidean (L2) distance is the most common, measuring the straight-line distance between two points in the N -dimensional space. However, in high-dimensional spaces, L2 distance can be sensitive to the magnitude of the vectors. Cosine similarity (or its inverse, Cosine distance) provides an alternative approach, measuring the cosine of the angle between two normalized feature vectors. This metric is mathematically defined as:

$$\cos(\theta) = \frac{A \cdot B}{\|A\| \times \|B\|} \quad (1)$$

For L2-normalized vectors ($\|A\| = \|B\| = 1$), this simplifies to the inner product $A \cdot B$. This makes cosine similarity invariant to vector magnitude, focusing purely on orientation [13]. Such direction-based comparisons often better correlate with semantic similarity, especially when features are L2-normalized [12].

For indexing, libraries such as FAISS (Facebook AI Similarity Search) [7] provide highly optimized implementations for both exact (brute-force) and approximate nearest-neighbor (ANN) search. FAISS supports multiple index types, including `IndexFlatL2` for exact L2 search and `IndexFlatIP` (Inner Product) for exact cosine similarity search on normalized vectors, as employed in this project. These index types are standard in similarity search literature and are well-documented in the FAISS library specifications.

3. Data

The CIFAR-10 dataset [6] was utilized to develop and evaluate the image retrieval framework. CIFAR-10 is a well-known benchmark dataset in computer vision, consisting of 60,000 color images distributed evenly across ten classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. Each image has a low resolution of 32×32 pixels and represents a single object.

For this study, a subset of 200 images (20 per class) was selected from the training set to enable efficient experimentation, indexing, and visualization while preserving class diversity. The 200-image subset was also used for the t-SNE visualization to provide a representative view of the embedding space.



Figure 1. Sample CIFAR-10 images. The grid shows one example from each of the ten classes used in the dataset.

All images were preprocessed before feature extraction. For the CNN-based methods, each 32×32 image was resized to 224×224 pixels to match the required input dimensions of the ResNet-50 architecture. Standard ImageNet normalization was then applied using channel-wise mean and standard deviation values:

$$\text{mean} = [0.485, 0.456, 0.406], \quad \text{std} = [0.229, 0.224, 0.225]. \quad (2)$$

This normalization ensures that the input distribution aligns with the data on which the model was originally trained.

For the Color Histogram baseline, images were also resized to 224×224 pixels to ensure a fair comparison but were not normalized, as the histogram algorithm operates directly on 0–255 pixel intensity values.

4. Methodology

4.1. Overview

This study implements a content-based image retrieval (CBIR) framework that uses pretrained convolutional neural network (CNN) embeddings for semantic similarity search. Deep features are extracted via a pretrained ResNet-50, and retrieval is performed using the FAISS (Facebook AI Similarity Search) library. Two distance metrics—Euclidean (L2) and Cosine similarity—are evaluated in an ablation study to determine which better preserves semantic relationships in the learned feature space.

The workflow comprises five stages: (1) dataset preparation, (2) feature extraction, (3) similarity indexing, (4) retrieval & visualization, and (5) performance evaluation. A traditional 3D Color Histogram baseline is implemented to highlight the contrast between handcrafted and deep representations. Qualitative clustering patterns are examined through t-Distributed Stochastic Neighbour Embedding (t-SNE) visualization.

4.2. Dataset Preparation

As outlined in Section 3, the CIFAR-10 dataset consists of 60,000 color images evenly distributed across ten object classes. For the retrieval experiments, a subset of 200 images (20 per class) was selected to balance class diversity

with computational efficiency, enabling rapid FAISS indexing and visualization.

Each image was resized from 32×32 to 224×224 pixels and normalized using standard ImageNet channel statistics:

$$\text{mean} = [0.485, 0.456, 0.406], \quad \text{std} = [0.229, 0.224, 0.225]. \quad (3)$$

This ensures compatibility with the pretrained ResNet-50 feature extractor. For the color-histogram baseline, images were resized but not normalized, since the method operates directly on 8-bit RGB intensity values. This preparation ensured that both retrieval pipelines—deep CNN and baseline histogram—used consistent input dimensions for a fair comparison.

4.3. Feature Extraction Using Pretrained CNN

The ResNet-50 architecture, pretrained on ImageNet, was employed as a fixed feature extractor. Instead of using its final classification layer, activations from the preceding average-pooling layer were taken as 2048-dimensional feature vectors (embeddings). Each embedding acts as a compact “fingerprint” encoding the object’s shape, texture, and color composition.

Two visually similar samples yield closely positioned embeddings, whereas dissimilar objects appear farther apart. These vectors form the searchable representation of all images in the dataset.

4.4. Similarity Measurement and Indexing

Image similarity was quantified by computing distances between embeddings in the feature space.

- **Euclidean (L2) distance:** Measures absolute magnitude differences but is sensitive to lighting or scale variations.
- **Cosine similarity:** Compares only vector orientation, making it invariant to magnitude and better suited to normalized embeddings.

Two FAISS indices were implemented [7]:

1. `IndexFlatL2` for Euclidean distance, and
2. `IndexFlatIP` for Cosine similarity (using L2-normalized embeddings, where the inner product $u \cdot v = \cos \theta$).

Both perform exhaustive nearest-neighbor searches to ensure comparability. Each index stores 2048-D vectors representing the dataset images.

During the query process, a given image—either from the dataset or an external source—undergoes identical preprocessing and feature extraction. The resulting vector is searched against the selected FAISS index to retrieve the



Figure 2. System architecture diagram. An input image is preprocessed (resized and normalized) and passed through ResNet-50 (with its classifier removed) to produce a 2048-D embedding. This embedding is compared against the FAISS index using L2 or Cosine similarity to return the most similar matches.

Top- K most similar items ($K = 5$ in this study). Retrieved images are displayed with similarity scores and class labels.

A retrieval threshold was applied to the cosine and histogram pipelines, ensuring that when similarity scores fell below a defined cutoff, the system reported “no visually similar images found,” thus preventing false matches for out-of-domain queries.

4.5. Baseline Comparison: Color Histogram

A traditional CBIR baseline using 3D Color Histograms was developed for quantitative contrast. Each RGB image was quantized into 8 bins per channel, producing a 512-dimensional ($8 \times 8 \times 8$) feature vector summarizing its color distribution. These histogram descriptors were indexed with FAISS (IndexFlatL2) and queried using Euclidean distance, ensuring identical retrieval logic to the CNN system.

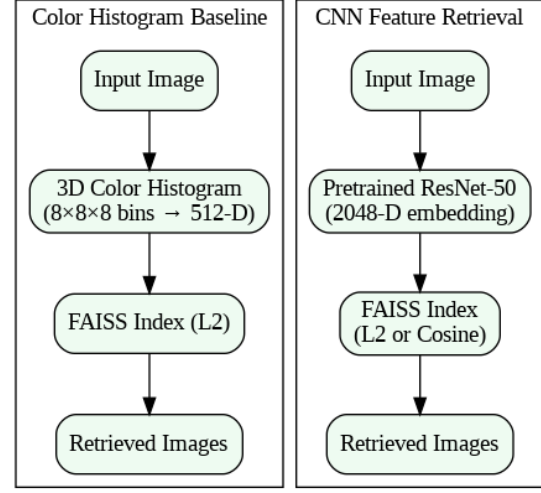
While computationally simple, this approach captures only low-level chromatic information. It cannot account for object shape, texture, or spatial context—two images sharing similar color tones (e.g., a blue car and a blue bird) may appear “close” despite differing semantically. Consequently, this baseline demonstrates the limits of handcrafted features and highlights the advantage of deep, learned representations.

4.6. Evaluation and Visualization

Precision@K Consistency: Retrieval performance was evaluated using Precision@10 ($P@10$), computed as the proportion of the top-10 retrieved images that share the majority class label among those results. This continuous metric (range 0–1) reflects how consistently the retrieved set aligns semantically with the query.

The implementation (`cnn_label_consistency`, `hist_label_consistency`) used $K = 10$ for quantitative evaluation, while the retrieval widgets displayed Top-5 images for visual clarity. This dual- K setup balances interpretability (Top-5 examples) with statistical reliability (Top-10 averaging).

Comparing L2 and Cosine versions quantitatively validated the ablation study (see Table 1). Cosine similarity consistently achieved higher $P@10$ scores, indicating more semantically accurate retrieval and supporting the hypothesis that direction-based similarity better captures class-level relationships in the embedding space [12].



Baseline vs. CNN Retrieval Pathways

Figure 3. Methodological comparison. Side-by-side depiction of two retrieval pipelines: (a) Baseline: Input Image $\rightarrow$ 3D Color Histogram ($8 \times 8 \times 8$ bins) $\rightarrow$ 512-D Vector $\rightarrow$ FAISS (L2) Search; (b) Proposed: Input Image $\rightarrow$ Pretrained ResNet-50 $\rightarrow$ 2048-D Embedding $\rightarrow$ FAISS (L2 or Cosine) Search.

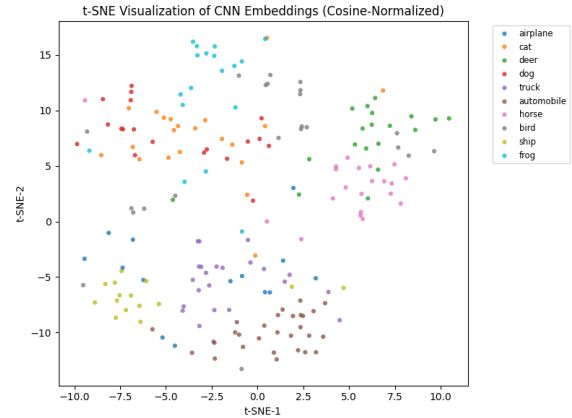


Figure 4. t-SNE visualization of 200 embeddings projected to two dimensions. Each point is color-coded by its true CIFAR-10 label. Clusters formed by cosine-normalized embeddings show clear class separation, confirming that direction-based similarity is more discriminative than magnitude.

Component	Implementation	Rationale
Backbone	ResNet-50 (ImageNet)	Proven transfer-learning architecture with robust hierarchical features.
Embedding Size	2048-D	High-capacity representation from penultimate layer.
Indexing Tool	FAISS (FlatL2 / FlatIP)	Enables fast, scalable nearest-neighbor search.
Baseline	3D Color Histogram (512-D)	Provides a classical non-deep comparison.
Similarity Metrics	L2 vs. Cosine (L2-normalized)	Tests direction- vs. magnitude-based retrieval.
Evaluation Metric	Precision@10	Measures semantic retrieval accuracy.
Visualization	t-SNE (200 embeddings)	Validates semantic clustering qualitatively.

Table 1. Summary of design choices adopted in this study.

4.7. Summary of Design Choices

5. Experiments and Results

5.1. Experimental Setup

The experiments were conducted using the `Image_Retrieval_CNN.ipynb` notebook. The core index was built from the 200-image CIFAR-10 subset. Three distinct retrieval systems were evaluated:

1. **CNN-L2:** ResNet-50 embeddings (2048-D) with `faiss.IndexFlatL2`.
2. **CNN-Cosine:** L2-normalized ResNet-50 embeddings (2048-D) with `faiss.IndexFlatIP`.
3. **Baseline-Hist:** 3D Color Histogram features (512-D) with `faiss.IndexFlatL2`.

The primary evaluation metric was Precision@10 (P@10), calculated using the `cnn_label_consistency` and `hist_label_consistency` functions. P@10 measures the proportion of the top-10 retrieved images that share the majority class label among those results. Top-5 retrievals followed the same trend (Cosine $\hat{>$ L2 $\hat{>$ Histogram), confirming that the performance pattern was consistent across K values.

5.2. Qualitative Retrieval Results

Figure 5 shows a qualitative comparison of the three methods using an *automobile* image as the query.

Analysis: CNN-Cosine retrieves semantically correct automobiles regardless of background or pose, showing strong semantic alignment. CNN-L2 also performs well but can be sensitive to vector magnitude, occasionally reducing robustness. Baseline-Hist retrieves images of similar color rather

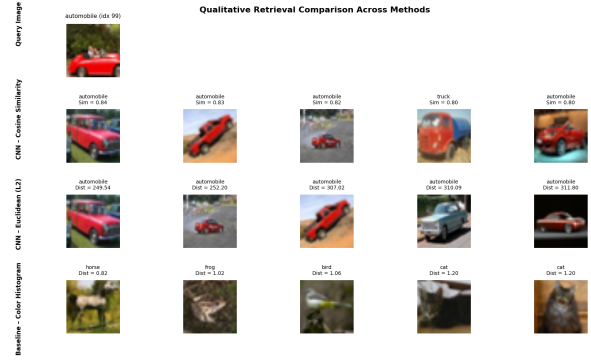


Figure 5. Qualitative retrieval results (Euclidean L2 vs. Cosine vs. Histogram). A 3×5 grid comparing retrieval outcomes for an “automobile” query. Row 1: query image. Row 2: CNN-Cosine results (high semantic consistency). Row 3: CNN-L2 results (accurate but magnitude-sensitive). Row 4: Color-Histogram results (color-based mismatches).

Method	Similarity Metric	P@10 (Avg.)
Baseline – Color Histogram	Euclidean (L2)	0.322
CNN (Proposed)	Euclidean (L2)	0.568
CNN (Ablation)	Cosine Similarity	0.598

Table 2. Ablation study: P@10 retrieval-consistency comparison.

than semantic class, exposing the classical “semantic gap” between pixel-level and concept-level similarity.

5.3. Ablation Study: Quantitative Results

To quantify the performance differences, P@10 label-consistency scores were computed for multiple queries. Results are aggregated in Table 2.

Analysis: The quantitative outcomes confirm the qualitative observations. The Color-Histogram baseline achieves only 32.2 % consistency, while the CNN-L2 method improves to 56.8 %. The CNN-Cosine variant yields the best result (59.8 %), verifying that angular similarity provides a more robust and semantically meaningful comparison of deep embeddings.

5.4. Visualization Analysis (t-SNE)

To further understand the superiority of Cosine similarity, a t-SNE visualization of 200 embeddings (20 per class) was generated, as shown in Figure 6.

Analysis: In Figure 6(a), raw ResNet-50 embeddings (L2 space) show weak separation. After L2-normalization, Figure 6(b) reveals clear, compact clusters, demonstrating that the orientation of embedding vectors carries the essential semantic information, whereas magnitude introduces noise. This visualization reinforces the conclusion that Cosine similarity better captures semantic alignment in high-dimensional feature spaces.

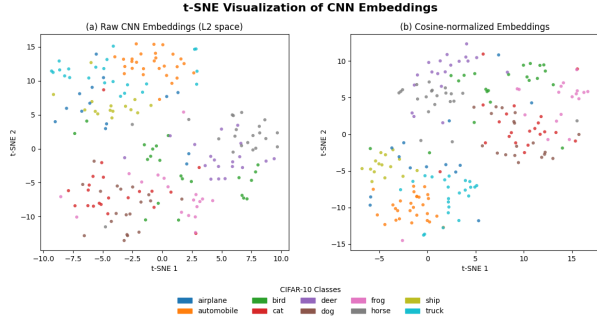


Figure 6. t-SNE visualization of 200 CIFAR-10 embeddings. (a) Unnormalized (L2) embeddings: overlapping, poorly separated clusters. (b) Cosine-normalized embeddings: tight, distinct clusters corresponding to the 10 CIFAR classes.

6. Conclusion and Future Work

6.1. Findings

This project successfully demonstrated the implementation of a high-performance, semantic image retrieval system using pretrained ResNet-50 embeddings and FAISS. The experiments confirmed that deep learning-based embeddings vastly outperform traditional low-level features like 3D Color Histograms, effectively bridging the semantic gap. The ablation study further revealed that for high-dimensional embeddings, Cosine similarity is a more effective metric than Euclidean (L2) distance—a result supported by both quantitative evidence (59.8% vs. 56.8% P@10 consistency) and qualitative t-SNE visualization.

6.2. Limitations

The primary limitation of this study lies in its scale. The retrieval index and visualization were built on only 200 images. While adequate for proof-of-concept and methodological comparison, performance may vary when applied to the full 60,000-image CIFAR-10 dataset or larger, more complex datasets. Additionally, only one CNN architecture (ResNet-50) and brute-force indexing (IndexFlat) were explored for exact search.

6.3. Future Work

This study lays the groundwork for several promising directions:

- **Scaling:** Extend the retrieval system to the complete CIFAR-10 dataset or to large-scale datasets such as COCO or ImageNet to evaluate scalability and performance.
- **Architecture Ablation:** Compare ResNet-50 with newer backbones such as Vision Transformers (ViT) or EfficientNet to analyze how representation differences influence retrieval quality.

- **Efficient Indexing:** For large-scale retrieval, replace brute-force IndexFlat with approximate nearest-neighbor (ANN) methods like FAISS’s IndexIVFPQ to achieve major speedups while maintaining high recall.
- **Interactive Application:** Expand the Jupyter-based prototype into a full web application, enabling users to perform real-time visual searches.

References

- [1] R. Szeliski, *Computer Vision: Algorithms and Applications*. Springer Science & Business Media, 2010.
- [2] M. J. Swain and D. H. Ballard, “Color indexing,” *International Journal of Computer Vision*, vol. 7, no. 1, pp. 11–32, 1991. DOI: 10.1007/BF00130487.
- [3] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, “ImageNet: A large-scale hierarchical image database,” in *Proc. IEEE CVPR*, 2009, pp. 248–255. DOI: 10.1109/CVPR.2009.5206848.
- [4] A. Krizhevsky, I. Sutskever, and G. E. Hinton, “ImageNet classification with deep convolutional neural networks,” *Communications of the ACM*, vol. 60, no. 6, pp. 84–90, 2017. DOI: 10.1145/3065386.
- [5] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE CVPR*, 2016, pp. 770–778. DOI: 10.1109/CVPR.2016.90.
- [6] A. Krizhevsky, “Learning multiple layers of features from tiny images,” University of Toronto, Technical Report, 2009.
- [7] J. Johnson, M. Douze, and H. Jégou, “Billion-scale similarity search with GPUs,” *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535–547, 2021. DOI: 10.1109/TB-DATA.2019.2921809.
- [8] L. van der Maaten and G. Hinton, “Visualizing data using t-SNE,” *Journal of Machine Learning Research*, vol. 9, pp. 2579–2605, 2008.
- [9] D. G. Lowe, “Distinctive image features from scale-invariant keypoints,” *International Journal of Computer Vision*, vol. 60, no. 2, pp. 91–110, 2004. DOI: 10.1023/B:VISI.0000029664.99615.94.
- [10] J. Sivic and A. Zisserman, “Video Google: A text retrieval approach to object matching in videos,” in *Proc. IEEE ICCV*, 2003, pp. 1470–1477. DOI: 10.1109/ICCV.2003.1238663.
- [11] K. Simonyan and A. Zisserman, “Very deep convolutional networks for large-scale image recognition,” in *Proc. ICLR*, 2015.

- [12] A. S. Razavian, H. Azizpour, J. Sullivan, and S. Carlsson, “CNN features off-the-shelf: An astounding baseline for recognition,” in *Proc. IEEE CVPRW*, 2014, pp. 512–519. DOI: 10.1109/CVPRW.2014.131.
- [13] F. Wang, W. Liu, H. Liu, and J. Cheng, “Deep metric learning with angular loss,” in *Proc. IEEE ICCV*, 2017, pp. 1866–1874. DOI: 10.1109/ICCV.2017.205.

Author Contributions

Harmandeep Pal: Implemented final code corrections, performed quantitative evaluation (P@10 analysis), and generated report visuals. Authored the Abstract, Introduction, Related Work, Results, and Conclusion sections.

Vedant Bhadri: Implemented CNN feature extraction (L2 and Cosine), FAISS indexing, Color Histogram baseline, and t-SNE visualization. Authored the Methodology section.

Both authors jointly edited and formatted the final manuscript.

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